

- Choose: **Dog** **Cat**
- Enter information: name, breed, age, body weight (one option for lb and one for kg).
if entered in lb, we need to convert into kg
- Select type of recipe: a. Conventional b. Grain-free c. BURF
- Select meat: group A Group B group C
- Select organ meat: (optional)
- Select grains (CHO): group A group B
- Select vegetables: group A group B
- Select oil:
- Select fruits:
- Select minerals:
- Others (optional):

Grain e
paste

A must include ingredients:

	Min 20 grams max 25 grams
egg shells or Calcium Carbonate	20 grams
yeast	1 grams
salt	100 grams
boiled egg (2 eggs)	15% of diet DM
liver (Beef, duck)	10 grams
kelp	10 g
cod-liver oil	10 g
fish oil	25 g
wheatgerm oil	10 g
bone meal	40 g
Oyster, canned	

column

Blood meal

For dog's convention recipe

Recipe Restrictions:

- Diet total mix must add up to 1000 g (**DM basis**) (column E row 62)
- Protein %: 32-40% (column C row 69)
- Fat%: >12% <17% (column C row 70)
- CHO%: >30 and < 45% (column C row 71)
- Fiber %: >3% and < 6% (column C row 73)
- Energy density (kcal/kg): 4000-4500 kcal/kg (column C row 68)
- Ca:P : > 1.4:1 and less than 2.1 (column C row 74)
- Omega-6:Omega-3: > 2 and < 6 (column C row 75)

Ingredients Restrictions:

Meat:

Give the option to choose at least one with a max of 2

- Total meat at least 20% and max 35% of diet DM (column E) → 20% - 35%.
- If the meat source has >30% fat (group B), a second meat source with the same meat type from group A must be used. The new protein (from group A) should be limited to less than 15% of diet DM (column E).
- When using meat from group C (either one or two meats) and if fat% didn't exceed 16% (column C row 70), meat C can be the only protein source. However, If the fat exceeds 16% (column C row 70), replace some meat C with a meat from group A until fat drops to <16% (column C row 70).
- When using meat from group A, if fat <12% (column C row 70), replace some of the meat A with meat from group B or C (use same meat type) until fat% is at least 13% (column C row 70).
- When you increase meat or any other ingredient and you want to lower total food mix (keep 1000 g; column E row 62), start by lowering grains (still keep grains restrictions though (not less than 30% of diet DM; column E)).
- If protein >40% (column C row 69), reduce meat (protein A) until protein is <39%, if the meat drops below 20% of diet DM (column E) and protein still > 40% (column C row 69), reduce liver in die mix but not less than 12% of diet DM (column E).

Organ meat:

Liver must be included in all diets at at least 12% of diet DM but not more than 15% of diet DM (column E)

Give the option to choose one or two organ meat (gizzard, tongue, kidney, heart) for a max of 5% of diet DM (column E)

Grains (CHO)

- Give the option to choose at least one with the max of two
- If using grains from Group B, you need to limit to 20% of diet DM (column E) and provide an option from group A to add. If using only grains from group A, group B is optional.
- Min 30% and max 40% (column E)
- When potato is selected, the min grain should 25% of diet DM (column E)
- When trying to lower protein % column C row 69, first increase grains without exceeding the upper limit of 40% (column E)

Vegetables

- Give the option to choose two (at least one) from group A. if only group A is chosen, a min 8% of diet DM (column E) and max is 15% will be used (column E)
- Group B use is optional, but may choose up to 2
- If group B is chosen, limit total group B to less than 7% of diet DM (column E) with the total of group A and B to not exceed 15% (column E)
- When potato is used (group C), limit total potatoes to <10% of diet DM and reduce grains (see before grains restrictions).

Oils:

- Fat % min is 12% and max 17% (column C row 70),
- Give options for coconut oil (keep at <10 grams in column E)
- When diet fat % is less than 12% (column C row 70), add 10 grams coconut oil
- if exceeds 17% (column C row 70), first adjust protein source (switch from meat B group to meat A group or from meat C group to meat A group). if fat% is still high (>17%), reduce all used oils by 10% each until oil drops below 17%. add another 10% drop for all oils if the first drop was not enough to lower fat below 17%.
- If turkey meat was used in the mix, you could add chicken meat to fix fat or protein %.
- Omega-6: omega-3 > 2 and <6 (column C row 75). If less than 2, remove fish oil and lower cod liver oil to 5 grams and add sunflower oil until ratio reaches 2. If > 6, increase fish oil until ratio drops below 5.

A
B
C

B
C
D

Fruits and others

- Fruits use is optional
- fruits keep $<2.5\%$ of diet and $>1.5\%$ (column E)
- if Pumpkin seeds or sunflower seeds are selected, keep at $<1.2\%$ of diet (Column E)

Fiber

- Fiber: >3 and $<6\%$. (column C row 73),
- If fiber is less than 3% (column C row 73),, first increase vegetables by 10% of diet DM (column E; without exceeding the upper limit of 15% mentioned before) but if fiber is still less than 3% , add psyllium husk or rice bran but don't exceed 4.5% fiber of diet DM
- If fiber $>3\%$, don't add psyllium husk or rice bran
- If fiber $>7\%$ (column C row 73),, reduce vegetables in the diet by 10% (column E) each until reaching fiber $<6\%$ (column C row 73),

Minerals:

- Give the option between egg shell and Calcium carbonate
- Ca% must be at least 1% and max 1.3% (column L Row 66)
- P% must be at least 0.5% (column O Row 66)
- Ca:P (column C row 74) should be >1.4 and less than 2 . If below 1.4 add more eggshell, or Calcium carbonate and if $>$ than 2 , reduce eggshells or Calcium carbonate
- If P is low, replace some of the Calcium carbonate (or egg shell) with Dicalcium phosphate or increase bone meal to no more than 15 grams (column E).